Crane Predictive Maintenance — KiCad Schematic & PCB Front-End

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This document contains the KiCad schematic and PCB design plan for the analog front-end for ADS131E08 with IEPE and bridge sensor support for crane predictive maintenance. It includes component placement, connectivity, and general design rules.

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1. Project Structure

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crane_pm_kicad/
├─ crane_pm_frontend.kicad_pro  # KiCad project file
├─ crane_pm_frontend.kicad_sch  # Schematic file
├─ crane_pm_frontend.kicad_pcb  # PCB layout file
├─ symbols/                     # Custom symbols if needed
├─ footprints/                  # Custom footprints
└─ libs/                        # Libraries
```

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2. Schematic Overview

2.1 Power Supply

- Input: 24VDC industrial supply
- LDO regulators: 5V for logic, 3.3V for ADC and op-amps
- Decoupling: 100 nF + 10 uF near all ICs

2.2 IEPE Front-End (per channel)

- LM334 current source for 4 mA excitation
- Series capacitor (1 uF) for AC coupling
- Differential amplifier (OPA2140 or similar)
- RC low-pass filter (fc ~10 kHz)
- Output to ADS131E08 INxP / INxN

2.3 Bridge Front-End (per channel)

- Bridge excitation: 2.5 V reference (REF5025 / ADR4525)
- INA333 instrumentation amplifier with gain 500
- Output low-pass filter (R=1k, C=180nF)

• Differential output to ADC

2.4 ADC Connections (ADS131E08)

- IN0P/N to IN7P/N: connect to front-end differential outputs
- DRDY -> MCU EXTI pin
- SPI lines: SCLK, MOSI, MISO, CS
- RESET / PWDN -> GPIO
- Reference: 3.3 V clean supply

2.5 Connectors

- J1: Power input 24V+, GND, Chassis
- J2: IEPE sensor channels (+/-/shield)
- J3: Bridge sensor channels (+/-/excitation/shield)
- J4: Digital comms (CAN, RS485, optional Ethernet)
- J5: Service/debug (SWD, UART, Boot0)

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3. PCB Layout Guidelines

- Partition analog (front-end + ADC) and digital zones (MCU + regulators)
- Star ground with single point at power entry
- Shielded connectors for IEPE and bridge sensors
- Short, matched differential traces to ADC inputs
- Guard rings around high impedance inputs (bridge/strain gauge)
- Decoupling capacitors near ICs (100 nF + 10 uF)
- TVS diodes and series resistors for sensor protection
- M12 industrial connectors for sensors, microSD connector for data logging

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4. Design Notes

- Ensure analog reference (Vref) is low-noise and clean
- Layout sensitive analog signals away from high-speed digital lines
- DRDY line triggers MCU DMA for precise sample timing
- Keep thermal paths for power ICs and regulators
- Silkscreen labels for each sensor channel and test points

Next Steps 1. Open KiCad project and draw schematic using above front-end components 2. Assign footprints to components (LM334, OPA2140, INA333, ADS131E08, REF5025) 3. Route PCB with analog/digital separation, differential pair routing, and star ground 4. Export Gerber files for fabrication

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Prepared for Pragvansh Dharap — Crane PM project — version 1.0